HDOC Day-38 Activity

Menu-Driven C Program Using String Functions

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Aim

To write a menu-driven C program using separate functions to perform different operations on a string.

Operations

1. Count spaces, digits, and special characters in a string.
2. Count the frequency of a given character in a string.
3. Toggle the case of each alphabetic character in a string.
4. Exit the program.

Concept Used

The program uses functions, strings, loops, switch-case, and character handling functions from **<ctype.h>**. Separate functions make the program easier to understand and reuse.

Algorithm

1. Start the program and display the menu.
2. Read the user's choice.
3. For choices 1 to 3, read a string using **fgets()**.
4. Choice 1: scan the string and count spaces, digits, and special characters.
5. Choice 2: read a character and count how many times it occurs in the string.
6. Choice 3: convert uppercase letters to lowercase and lowercase letters to uppercase.
7. Choice 4: terminate the program.
8. For an invalid choice, display an error message.
9. Repeat until Exit is selected.

C Program

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

void countCharacters(char str[])
{
    int spaces = 0, digits = 0, special = 0, i;

    for (i = 0; str[i] != '\0'; i++)
    {
        if (str[i] == ' ')
            spaces++;
        else if (isdigit((unsigned char)str[i]))
            digits++;
        else if (!isalpha((unsigned char)str[i]) && str[i] != '\n')
            special++;
```

```
    }

    printf("Spaces = %d\n", spaces);
    printf("Digits = %d\n", digits);
    printf("Special characters = %d\n", special);
}

void countFrequency(char str[])
{
    char ch;
    int count = 0, i;

    printf("Enter character to find: ");
    scanf(" %c", &ch);

    for (i = 0; str[i] != '\0'; i++)
    {
        if (str[i] == ch)
            count++;
    }

    printf("Frequency of '%c' = %d\n", ch, count);
}

void toggleCase(char str[])
{
    int i;

    for (i = 0; str[i] != '\0'; i++)
    {
        if (islower((unsigned char)str[i]))
            str[i] = toupper((unsigned char)str[i]);
        else if (isupper((unsigned char)str[i]))
            str[i] = tolower((unsigned char)str[i]);
    }

    printf("String after toggling case: %s", str);
}

int main()
{
    char str[200];
    int choice;

    do
    {
        printf("\n----- MENU -----\n");
        printf("1. Count spaces, digits and special characters\n");
        printf("2. Count frequency of a character\n");
        printf("3. Toggle case of each character\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
        getchar();

        switch (choice)
        {
            case 1:
                printf("Enter a string: ");
                fgets(str, sizeof(str), stdin);
                countCharacters(str);
                break;

            case 2:
                printf("Enter a string: ");
                fgets(str, sizeof(str), stdin);
                countFrequency(str);
                getchar();
                break;

            case 3:
                printf("Enter a string: ");
                fgets(str, sizeof(str), stdin);
```

```
                toggleCase(str);
                break;

            case 4:
                printf("Exiting program...\n");
                break;

            default:
                printf("Invalid choice! Please try again.\n");
        }

    } while (choice != 4);

    return 0;
}
```

Sample Output

```
----- MENU -----
1. Count spaces, digits and special characters
2. Count frequency of a character
3. Toggle case of each character
4. Exit
Enter your choice: 1
Enter a string: Hello 123 @ C
Spaces = 3
Digits = 3
Special characters = 1

Enter your choice: 2
Enter a string: programming
Enter character to find: m
Frequency of 'm' = 2

Enter your choice: 3
Enter a string: Hello World
String after toggling case: hELLO wORLD

Enter your choice: 4
Exiting program...
```

Function Explanation

countCharacters(): Traverses the string and counts blank spaces, numeric digits, and characters that are neither letters nor digits.

countFrequency(): Takes a character from the user and counts its occurrences in the given string.

toggleCase(): Uses **islower()**, **isupper()**, **toupper()**, and **tolower()** to reverse the case of alphabetic characters.

main(): Displays the menu, accepts the user's choice, reads the string, and calls the required function using a switch statement.

Result

Thus, a menu-driven C program was successfully implemented using separate functions to count spaces/digits/special characters, find character frequency, toggle character case, and exit the program.

Quick Viva Questions

1. Why is <ctype.h> used? It provides character-checking and case-conversion functions.
2. Why are separate functions used? They make the program modular, readable, and reusable.
3. Why is fgets() used? It can read a complete line including spaces.
4. What does isdigit() do? It checks whether a character is a digit from 0 to 9.
5. What is case toggling? It changes uppercase letters to lowercase and lowercase letters to uppercase.